



Deep Learning Enhanced SLAM I: Pose Estimation and Relocalization

Graduate Course INTR-6000P

Week 11 - Lecture 21

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Recap: Planning for Embodied AI

Definition: Finding a sequence of actions to achieve a goal given constraints and a model of the world.

Analogy: “Navigation GPS for robots.”

Planning vs. Decision-making vs. Control

- Planning: *Strategic* (where to go)
- Control: *Tactical* (how to move)
- Decision-making: *High-level* (what to achieve)



Recap: Planning for Robotics

Popular Approaches

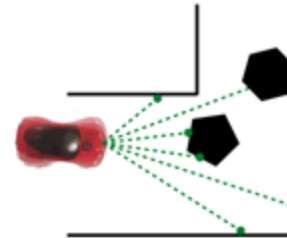
- **Potential fields** [*Rimon, Koditschek, '92*]: create forces on the robot that pull it toward the goal and push it away from obstacles
- **Grid-based planning** [*Stentz, '94*]: discretizes problem into grid and runs a graph-search algorithm (Dijkstra, A*, ...)
- **Combinatorial planning** [*LaValle, '06*]: constructs structures in the configuration space (C-space) that completely capture all information needed for planning
- **Sampling-based planning** [*Kavraki et al, '96; LaValle, Kuffner, '06, etc.*]: uses collision detection algorithms to probe and incrementally search the C-space for a solution

Recap: Robotic Control

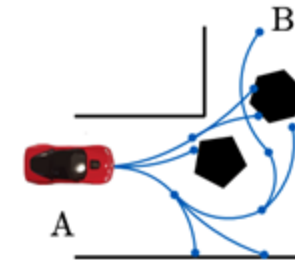
What Is Control?

- Definition: **Ensuring the robot follows planned motion despite disturbances and model errors.**
- Role in embodied AI: Bridge between planning and actuation.

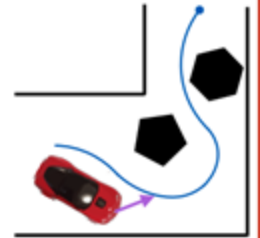
Estimate state



Plan a sequence of motions



Control robot to follow path



- Robot pose known
- Path is given

Recap: PID Controller

Control input $u(t)$ is proportional to **error**, its **accumulation**, and its **rate of change**.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Where:

- $e(t) = r(t) - y(t)$: tracking error
- K_p : proportional gain
- K_i : integral gain
- K_d : derivative gain

Recap: MPC

Model Predictive Control (MPC)

At each time step, **predict future behavior** using a system model and **optimize future control inputs** to minimize a cost function.

$$\min_{u_{0:H}} \sum_{k=0}^H \left[(x_k - x_{\text{goal}})^T Q (x_k - x_{\text{goal}}) + u_k^T R u_k \right]$$

- Q : penalty for state error (tracking error)
- R : penalty for control effort
- Balances **accuracy vs smooth control**

MPC finds control inputs that track target while minimizing effort and respecting constraints.

Lecture Outline

Part I

1. Motivation for Deep Learning in SLAM
2. End-to-End Pose Estimation
3. Learning based Camera Relocalization

Part II

1. Differentiable SLAM Back-End
2. Semantic SLAM

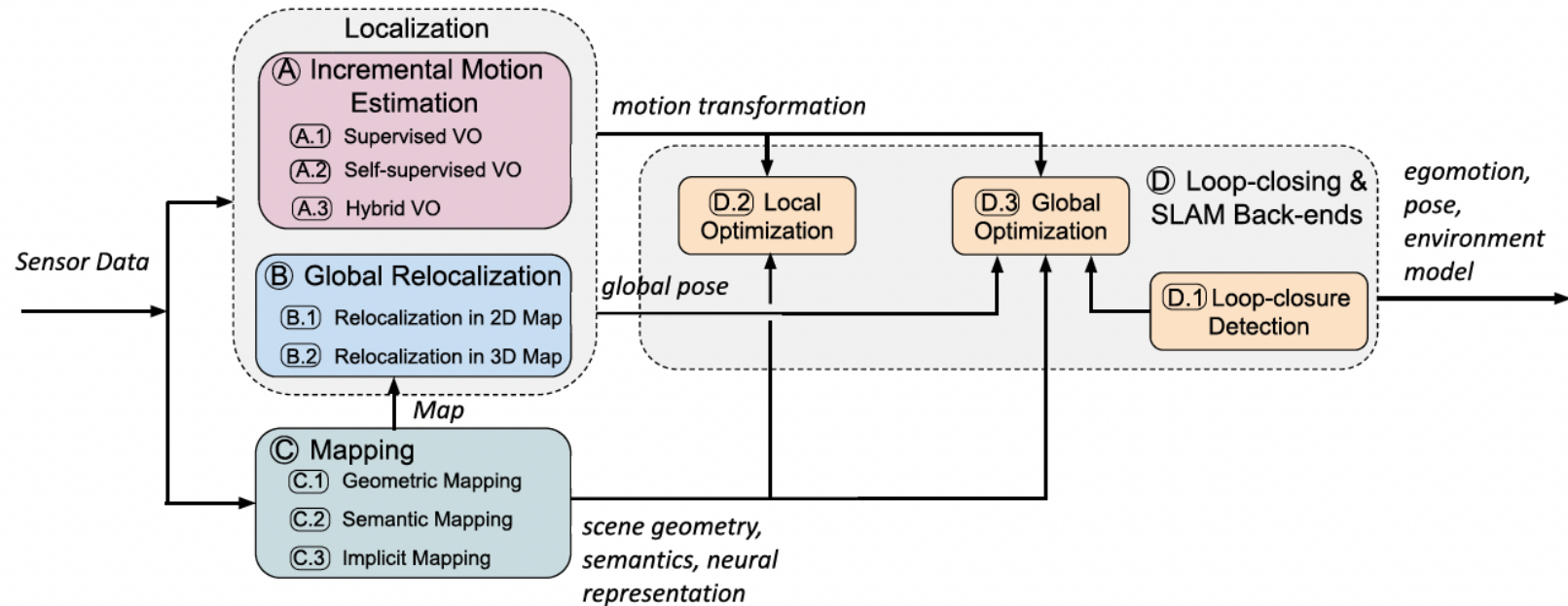
Deep Learning in SLAM

Challenges of classical SLAM:

- Feature sparsity in low-texture/low-light environments
- Scale ambiguity in monocular SLAM
- Difficulty in robust relocalization

Deep learning opportunities:

- End-to-end pose regression
- Learned features robust to illumination changes
- Semantic understanding for scene consistency



End-to-End Pose Estimation

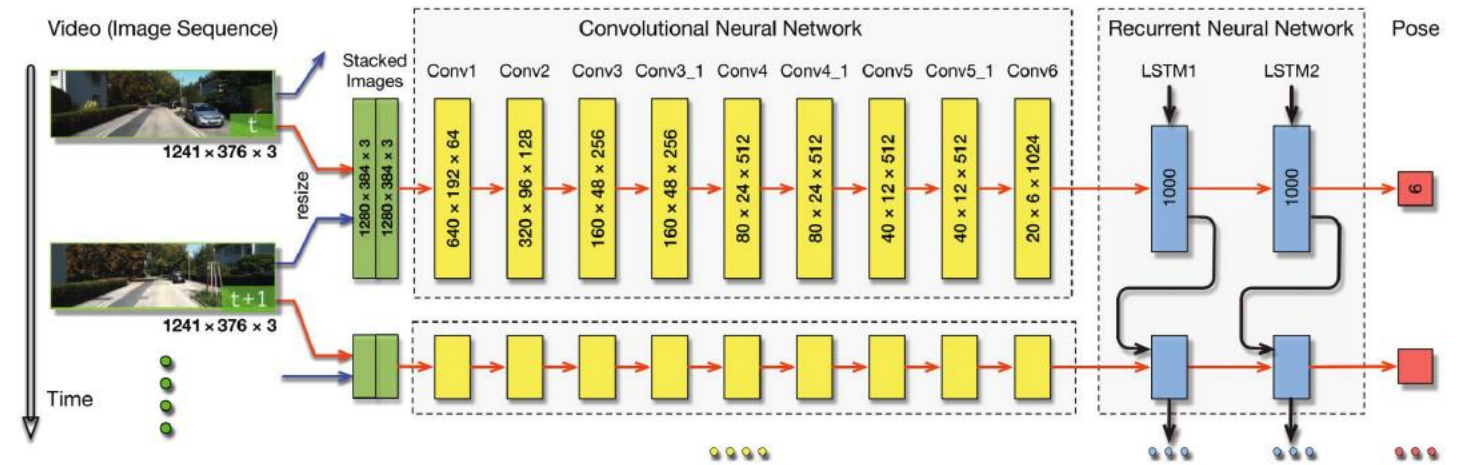
Problem: Predict relative camera pose directly from images

Methods:

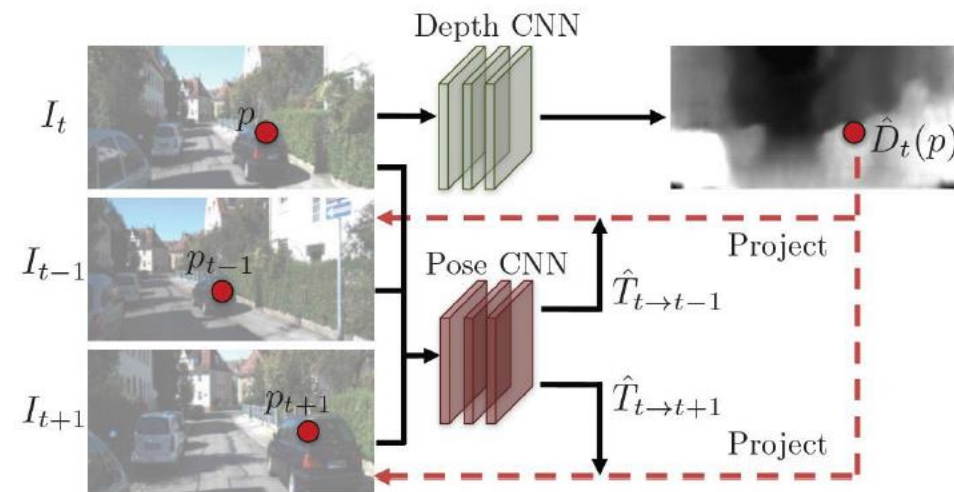
- Self-Supervised VO
- Supervised VO
- Hybrid VO

Input: Single/multiple frames or sequences

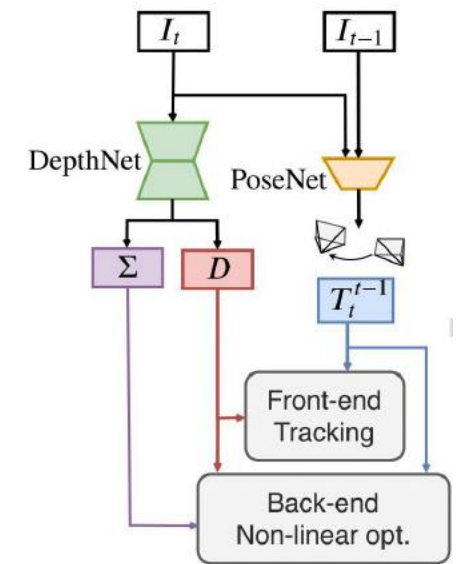
Output: 6-DoF pose



(a)



(b)



(c)

End-to-End Pose Estimation

Supervised learning for Visual Odometry

DeepVO: Towards End-to-End Visual Odometry with Deep Recurrent Convolutional Neural Networks (2017)

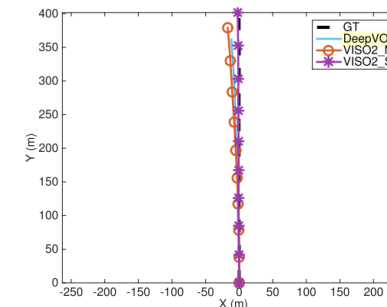
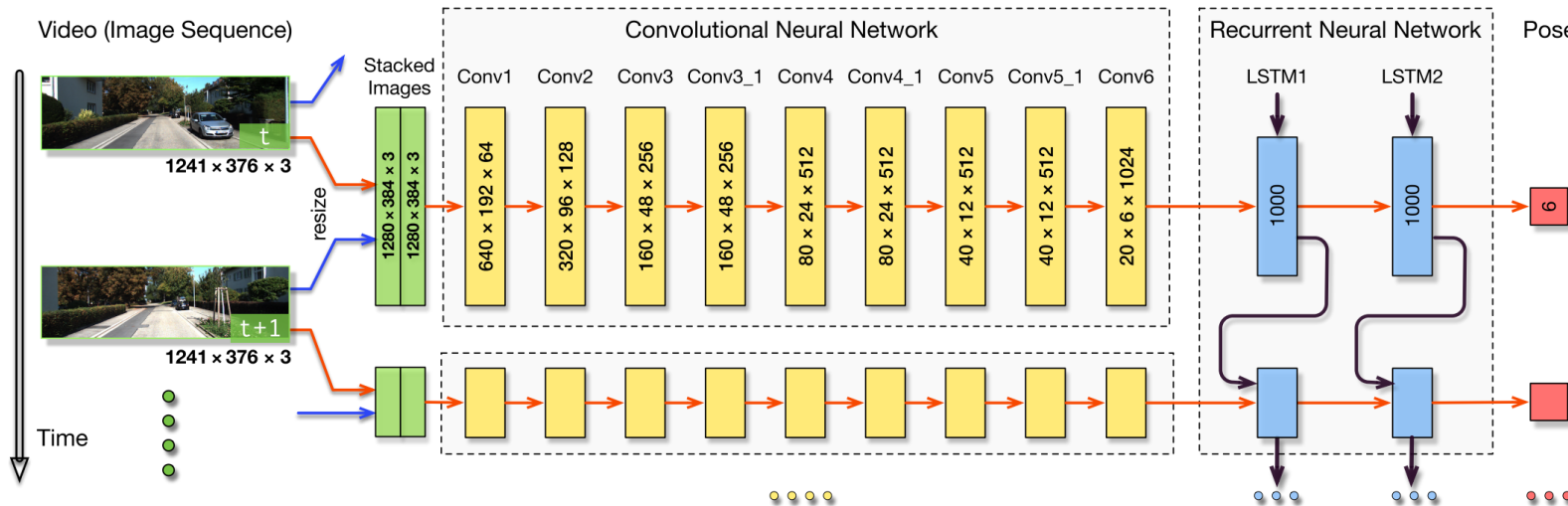
- CNN + LSTM
- Sequential modelling
- Works with monocular input

Advantages

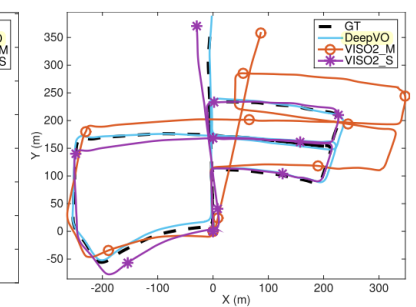
- End-to-end learning: Learns feature extraction + motion estimation jointly, without handcrafted features.
- Robust to visual degradation
- No need for camera calibration or explicit geometry

Limitations:

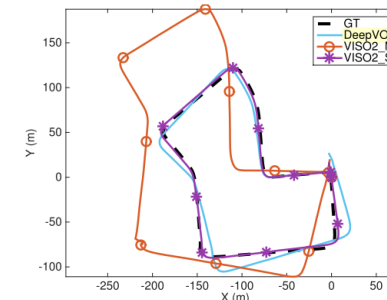
- Not physically interpretable
- Requires large labeled datasets
- Fails in unseen environments



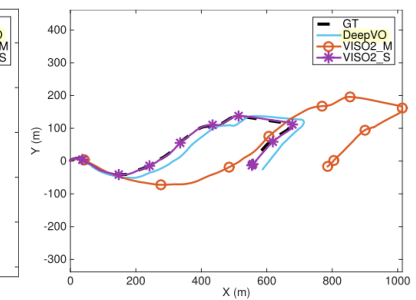
(a) Sequence 04.



(b) Sequence 05.



(c) Sequence 07.



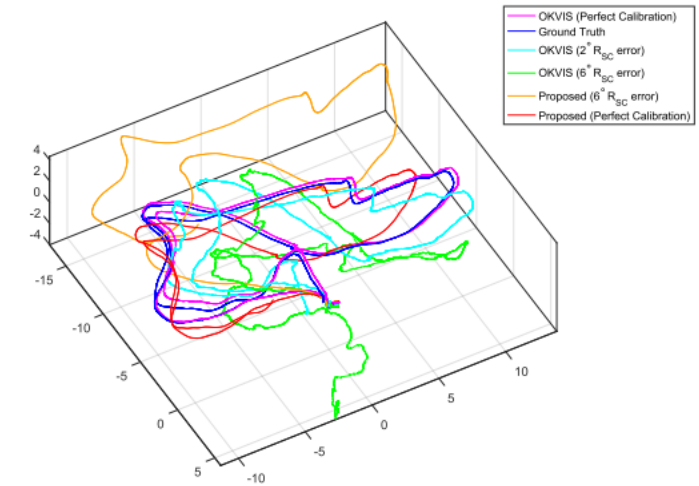
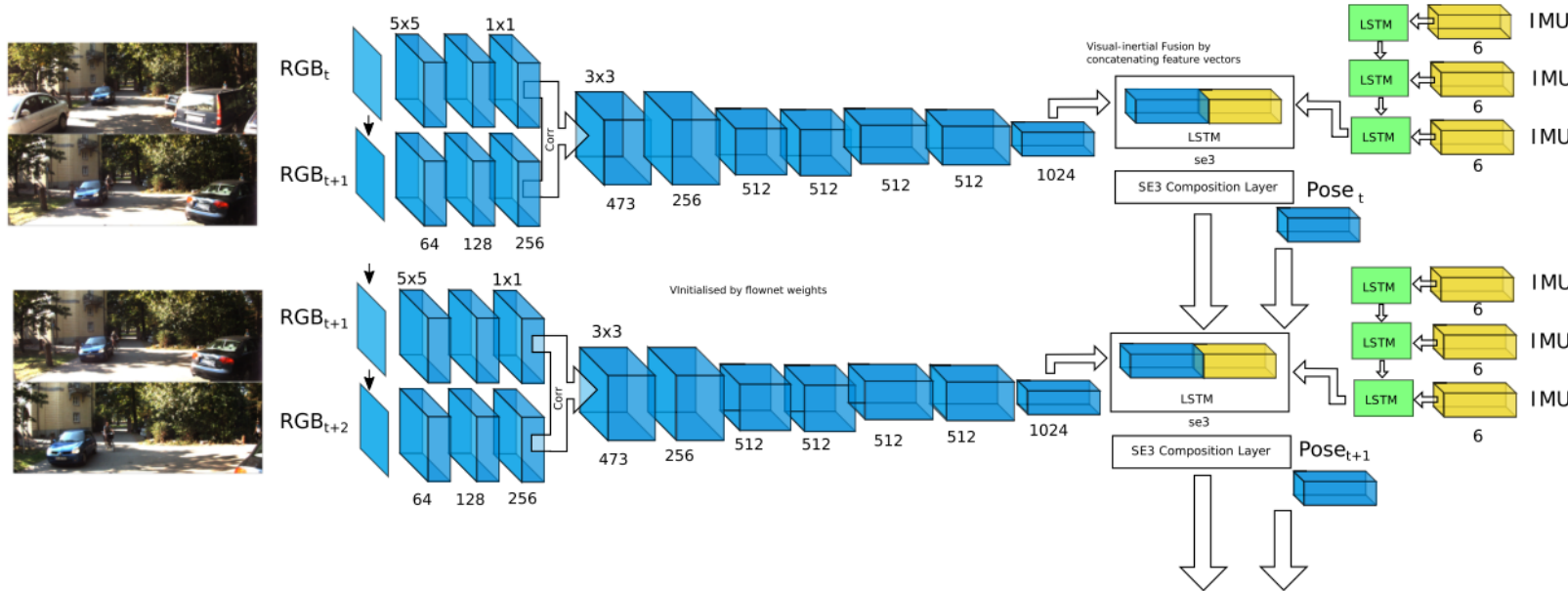
(d) Sequence 10.

End-to-End Pose Estimation

Supervised learning for Visual Odometry

VINet: Visual-inertial odometry as a sequence-to-sequence learning problem (2017)

- Fusion of IMU + images in RNN
- Multi-rate sensor fusion.
- More robust to sensor noises
- Can not compete with SoTA classical Visual-inertial odometry



End-to-End Pose Estimation

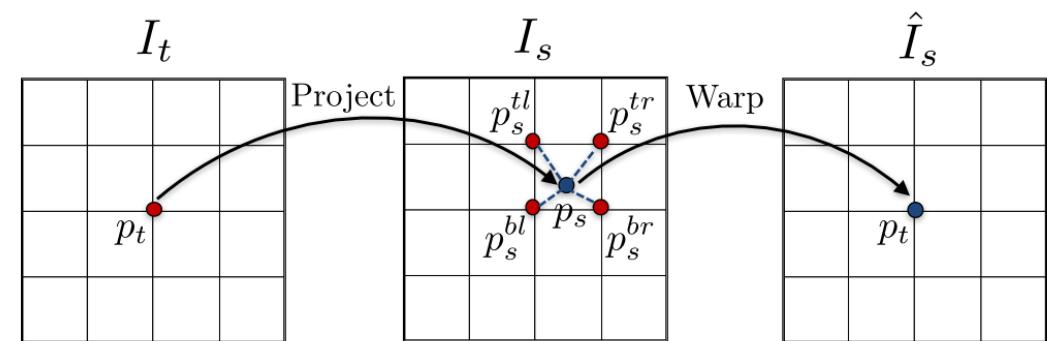
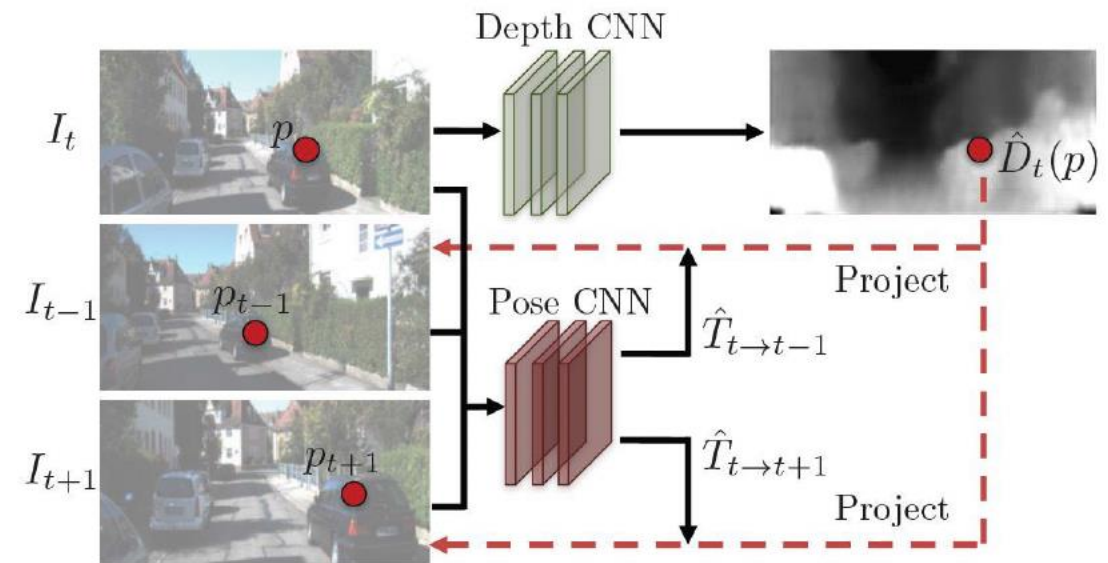
Self-supervised learning for Visual Odometry

Unsupervised Learning of Depth and Ego-Motion from Video (2017)

- Using view synthesis as supervision
- Pose + depth estimation jointly

Let p_t denote the homogeneous coordinates of a pixel in the target view, and K denote the camera intrinsics matrix. We can obtain p_t 's projected coordinates onto the source view p_s by²

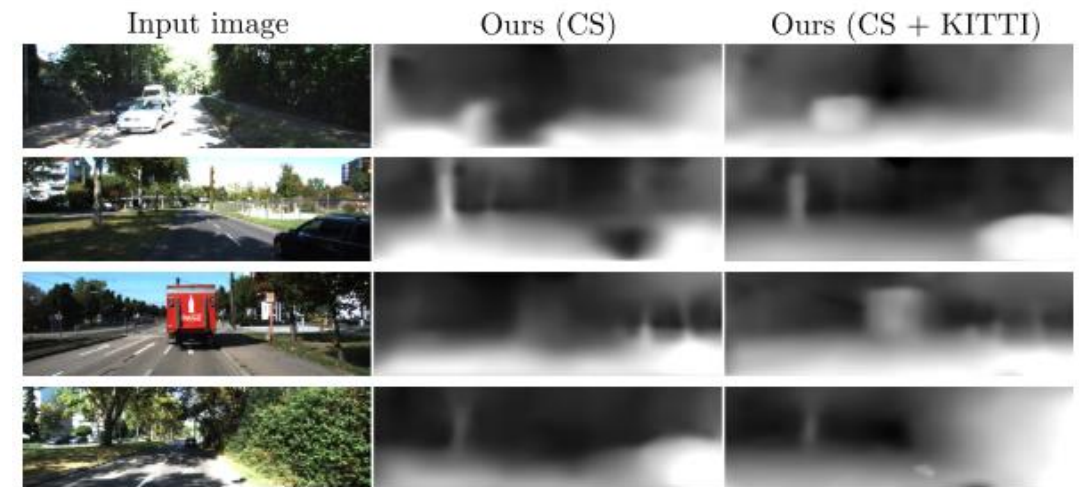
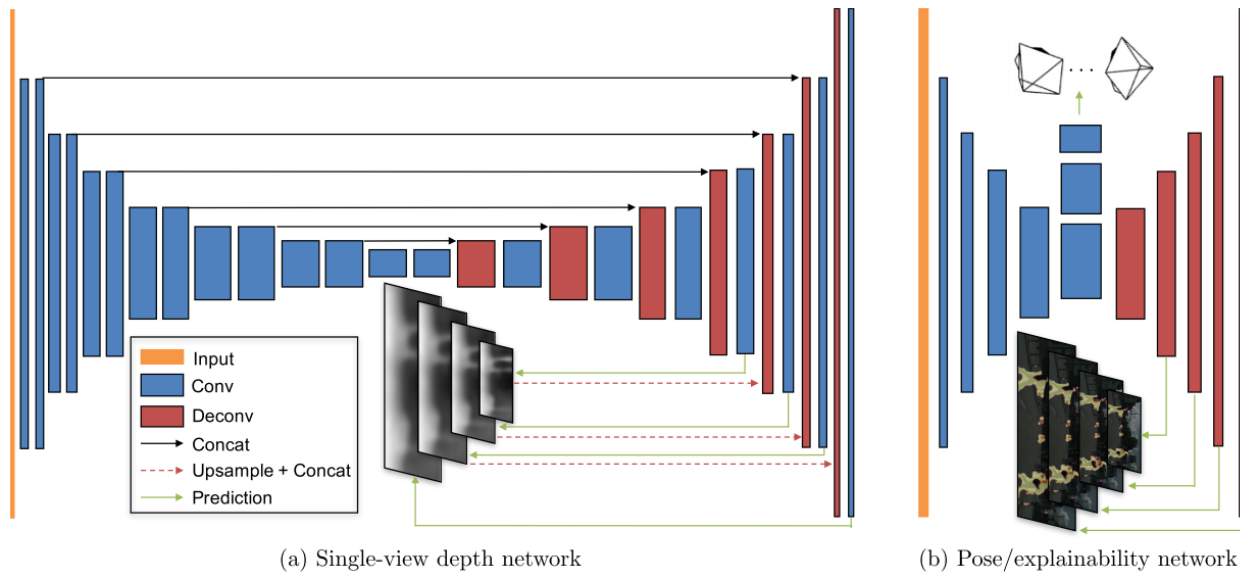
$$p_s \sim K \hat{T}_{t \rightarrow s} \hat{D}_t(p_t) K^{-1} p_t \quad (2)$$



End-to-End Pose Estimation

Self-supervised learning for Visual Odometry

Unsupervised Learning of Depth and Ego-Motion from Video (2017)



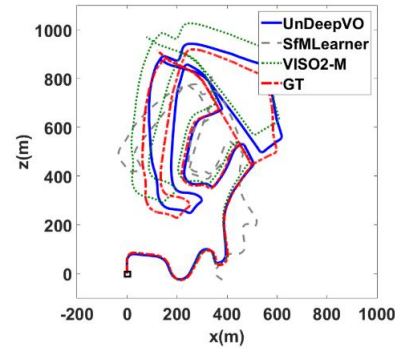
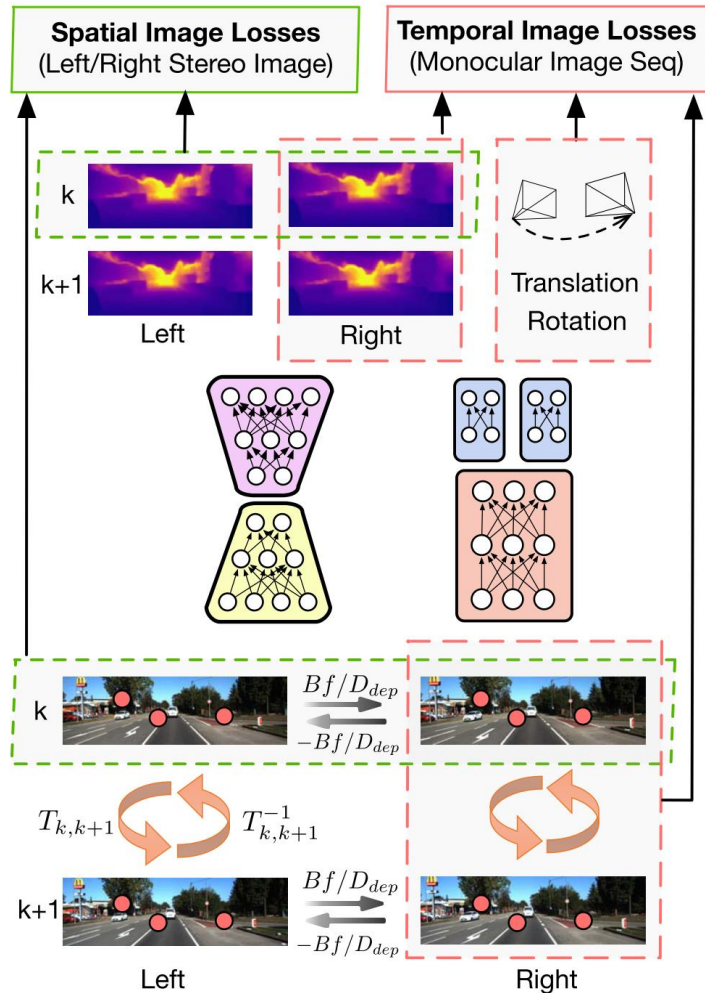
Advantages: No need for precise ground truth

Limitations: Scale drift, less interpretability

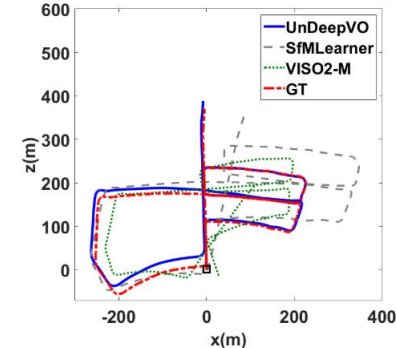
End-to-End Pose Estimation

Self-supervised learning for Visual Odometry

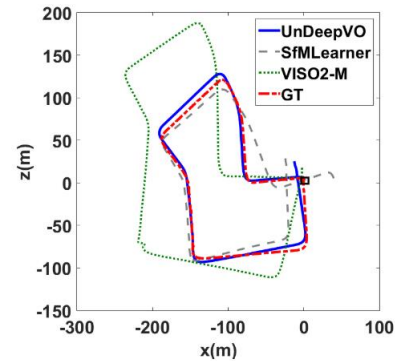
UnDeepVO: Monocular Visual Odometry Through Unsupervised Deep Learning (2018)



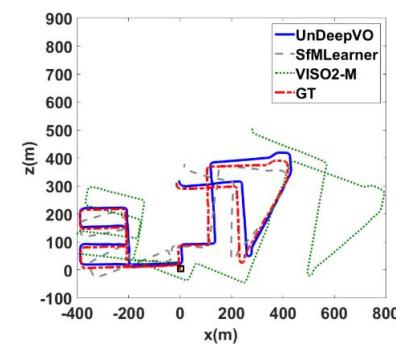
(a) 02



(b) 05



(c) 07



(d) 08

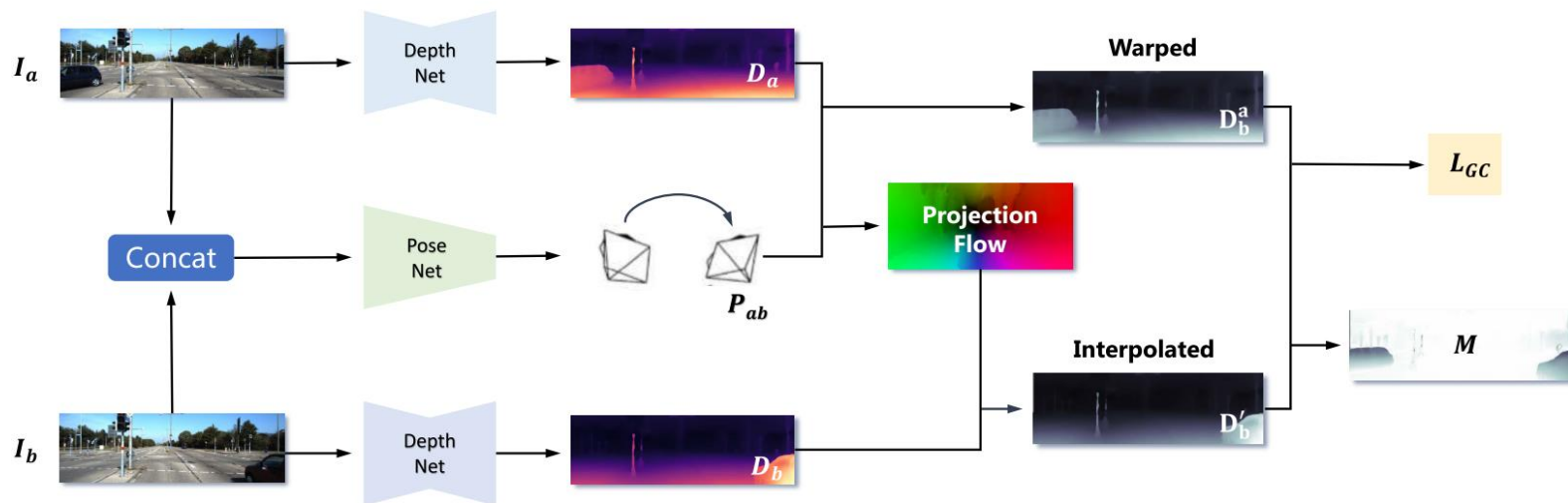
$$L_{dis}^l = L^{l_1}(D_{dis}^l, D_{dis}^{l'})$$

$$L_{dis}^r = L^{l_1}(D_{dis}^r, D_{dis}^{r'})$$

End-to-End Pose Estimation

Self-supervised learning for Visual Odometry

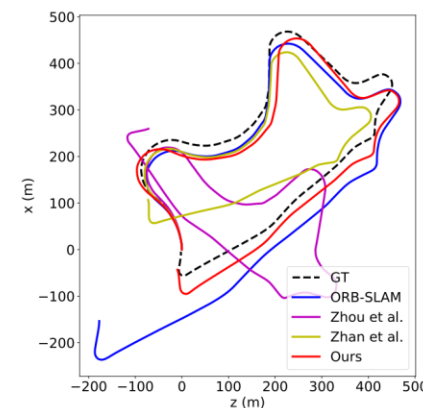
Unsupervised scale-consistent depth and ego-motion learning from monocular video (2019)



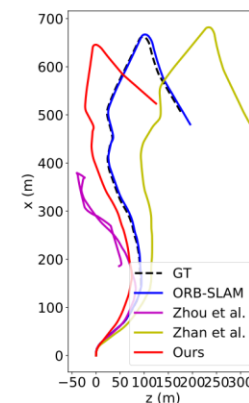
$$D_{\text{diff}}(p) = \frac{|D_b^a(p) - D_b^b(p)|}{D_b^a(p) + D_b^b(p)}$$

$$L_{GC} = \frac{1}{|V|} \sum_{p \in V} D_{\text{diff}}(p),$$

Methods	Seq. 09		Seq. 10	
	t_{err} (%)	r_{err} ($^{\circ}/100m$)	t_{err} (%)	r_{err} ($^{\circ}/100m$)
ORB-SLAM [12]	15.30	0.26	3.68	0.48
Zhou et al. [7]	17.84	6.78	37.91	17.78
Zhan et al. [17]	11.93	3.91	12.45	3.46
Ours (K)	11.2	3.35	10.1	4.96
Ours (CS+K)	8.24	2.19	10.7	4.58



(a) sequence 09

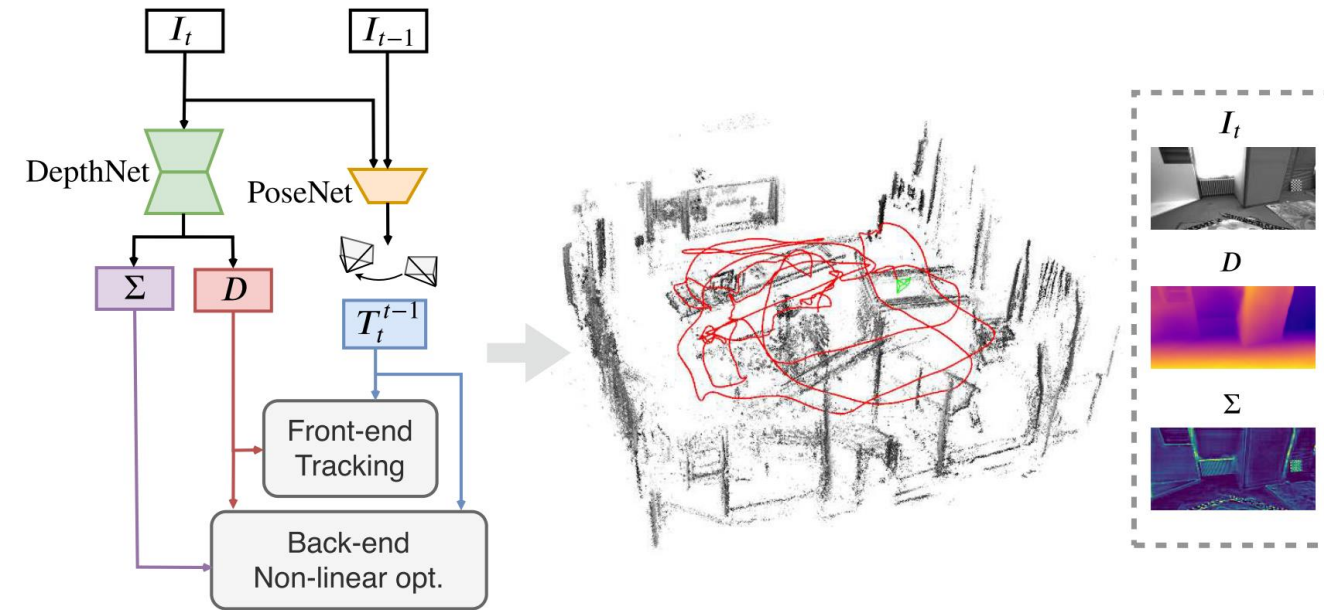


(b) sequence 10

End-to-End Pose Estimation

Hybrid Approach for Visual Odometry

D3VO: Deep Depth, Deep Pose and Deep Uncertainty for Monocular Visual Odometry (2020)



Main Components:

•Deep Depth Prediction

- Monocular depth estimated by a self-supervised network
- Provides accurate scene geometry for photometric VO

•Deep Pose Prior

- Network predicts relative pose between frames
- Used as an initialization for optimization
→ improves convergence

•Learned Uncertainty Estimation

- Pixel-wise uncertainty maps
- Down-weights unreliable image regions during optimization (e.g., occlusion, textureless areas)

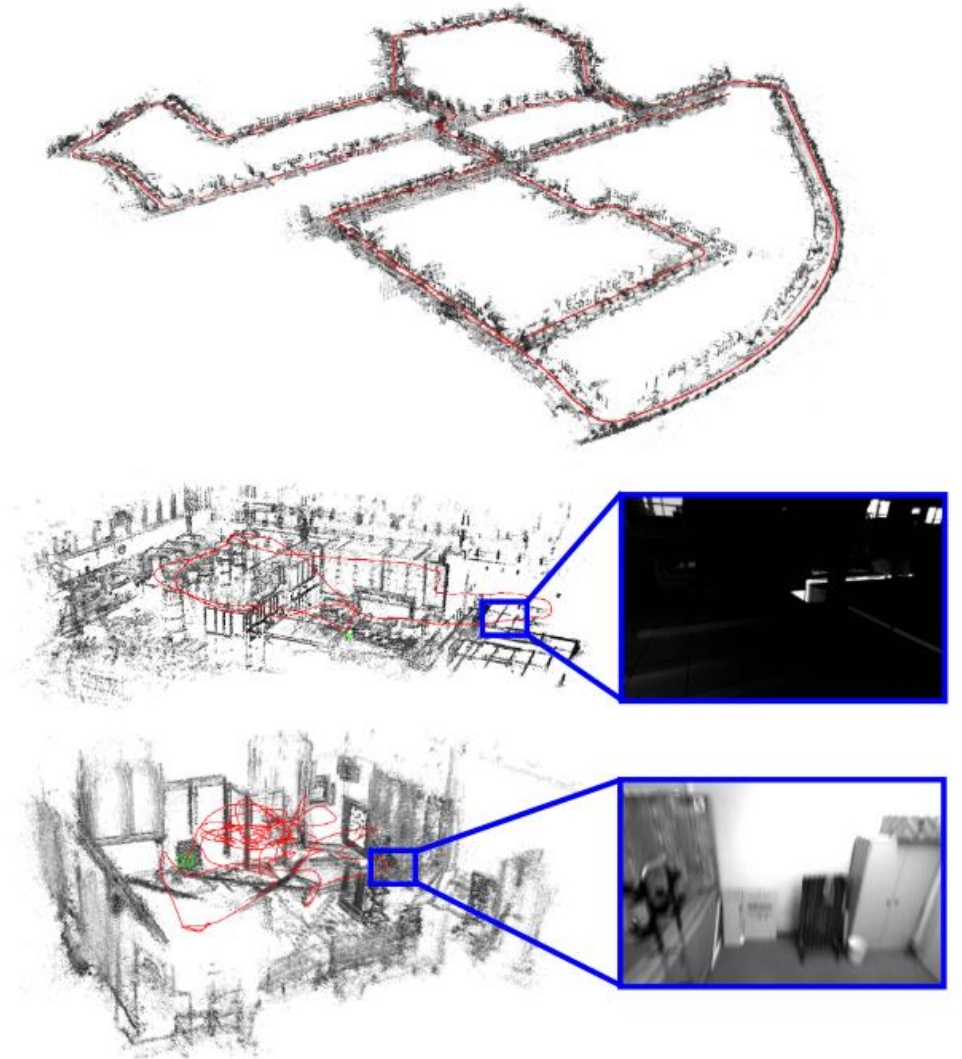
End-to-End Pose Estimation

Hybrid Approach for Visual Odometry

D3VO: Deep Depth, Deep Pose and Deep Uncertainty for Monocular Visual Odometry (2020)

		Seq. 09	Seq. 10
End-to-end	UnDeepVO [43]	7.01	10.63
	SfMLearner [81]	17.84	37.91
	Zhan et al. [78]	11.92	12.45
	Struct2Depth [5]	10.2	28.9
	Bian et al. [1]	11.2	10.1
	SGANVO [19]	4.95	5.89
	Gordon et al. [26]	2.7	6.8
Hybrid	CNN-SVO [44]	10.69	4.84
	Yin et al. [76]	4.14	1.70
	Zhan et al. [79]	2.61	2.29
	DVSO [74]	0.83	0.74
	D³VO	0.78	0.62

Table 5: Comparison to other hybrid methods as well as end-to-end methods on Seq.09 and 10 of KITTI Odometry.



Learning based Camera Relocalization

End-to-End Camera Relocalization

PoseNet: A Convolutional Network for Real-Time 6-DOF Camera Relocalization (2015)

- CNN architecture
- Regression losses

$$\mathbf{p} = [\mathbf{x}, \mathbf{q}]$$

- Quaternion vs Euler
- Scale balancing
- Geometric consistency losses.

$$loss(I) = \|\hat{\mathbf{x}} - \mathbf{x}\|_2 + \beta \left\| \hat{\mathbf{q}} - \frac{\mathbf{q}}{\|\mathbf{q}\|} \right\|_2$$

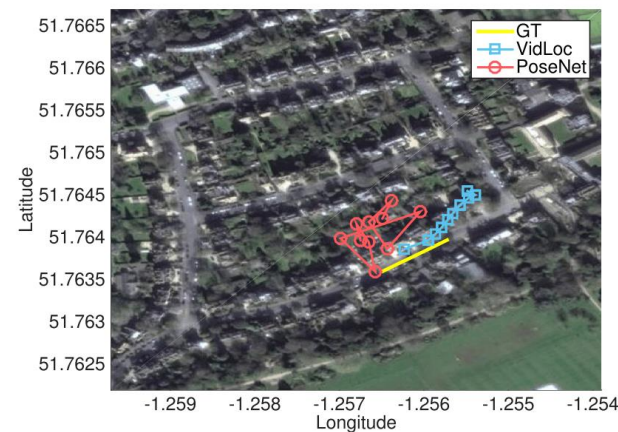
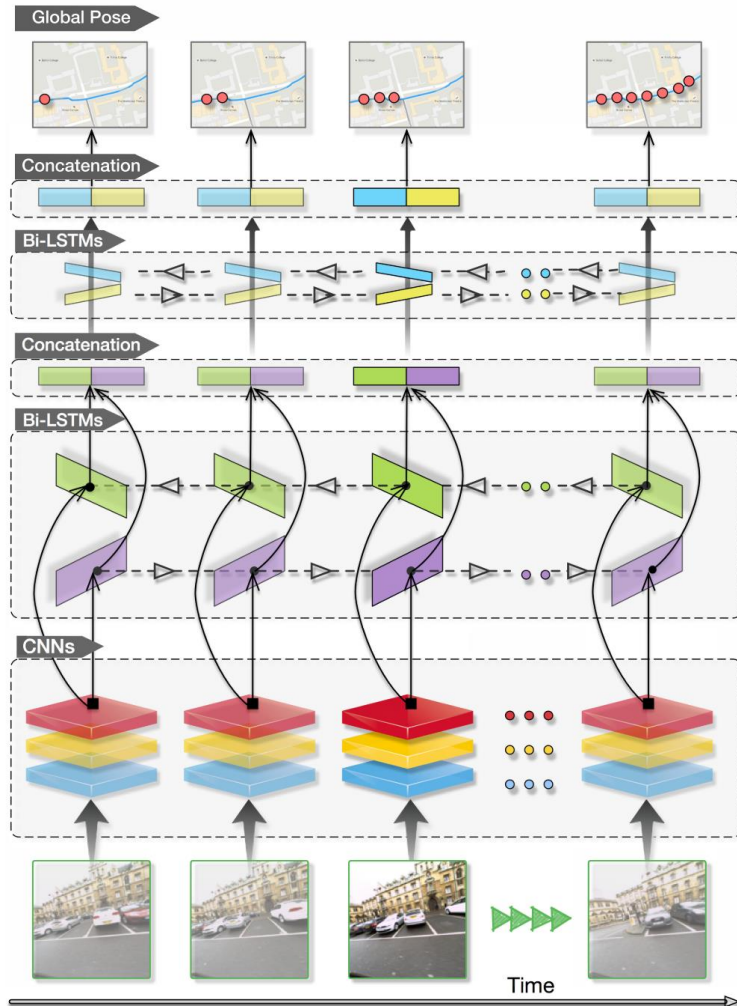
Figure 5: **7 Scenes dataset** example images from left to right; Chess, Fire, Heads, Office, Pumpkin, Red Kitchen and Stairs.

Scene	# Frames		Spatial Extent (m)	SCoRe Forest (Uses RGB-D)	Dist. to Conv. Nearest Neighbour	PoseNet	Dense PoseNet
	Train	Test					
King's College	1220	343	140 x 40m	N/A	3.34m, 2.96°	1.92m, 2.70°	1.66m, 2.43°
Street	3015	2923	500 x 100m	N/A	1.95m, 4.51°	3.67m, 3.25°	2.96m, 3.00°
Old Hospital	895	182	50 x 40m	N/A	5.38m, 4.51°	2.31m, 2.69°	2.62m, 2.45°
Shop Façade	231	103	35 x 25m	N/A	2.10m, 5.20°	1.46m, 4.04°	1.41m, 3.59°
St Mary's Church	1487	530	80 x 60m	N/A	4.48m, 5.65°	2.65m, 4.24°	2.45m, 3.98°
Chess	4000	2000	3 x 2 x 1m	0.03m, 0.66°	0.41m, 5.60°	0.32m, 4.06°	0.32m, 3.30°
Fire	2000	2000	2.5 x 1 x 1m	0.05m, 1.50°	0.54m, 7.77°	0.47m, 7.33°	0.47m, 7.02°
Heads	1000	1000	2 x 0.5 x 1m	0.06m, 5.50°	0.28m, 7.00°	0.29m, 6.00°	0.30m, 6.09°
Office	6000	4000	2.5 x 2 x 1.5m	0.04m, 0.78°	0.49m, 6.02°	0.48m, 3.84°	0.48m, 3.62°
Pumpkin	4000	2000	2.5 x 2 x 1m	0.04m, 0.68°	0.58m, 6.08°	0.47m, 4.21°	0.49m, 4.06°
Red Kitchen	7000	5000	4 x 3 x 1.5m	0.04m, 0.76°	0.58m, 5.65°	0.59m, 4.32°	0.58m, 4.17°
Stairs	2000	1000	2.5 x 2 x 1.5m	0.32m, 1.32°	0.56m, 7.71°	0.47m, 6.93°	0.48m, 6.54°

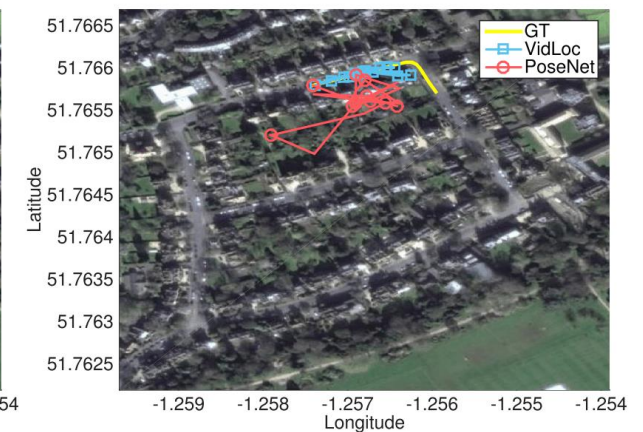
Learning based Camera Relocalization

End-to-End Camera Relocalization

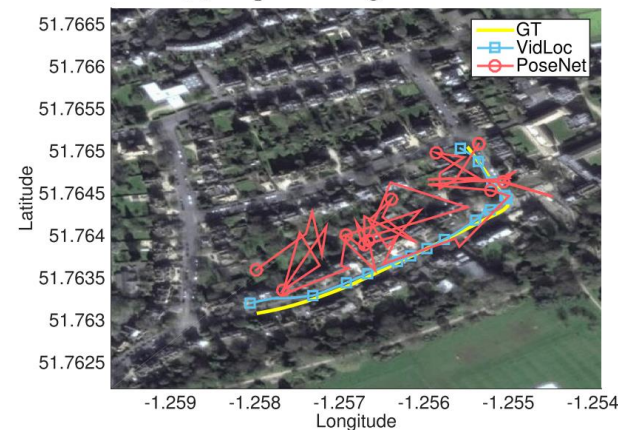
VidLoc: A Deep Spatio-Temporal Model for 6-DoF Video-Clip Relocalization (2017)



(a) Sequence length: 10.



(b) Sequence length: 20.



(c) Sequence length: 50.

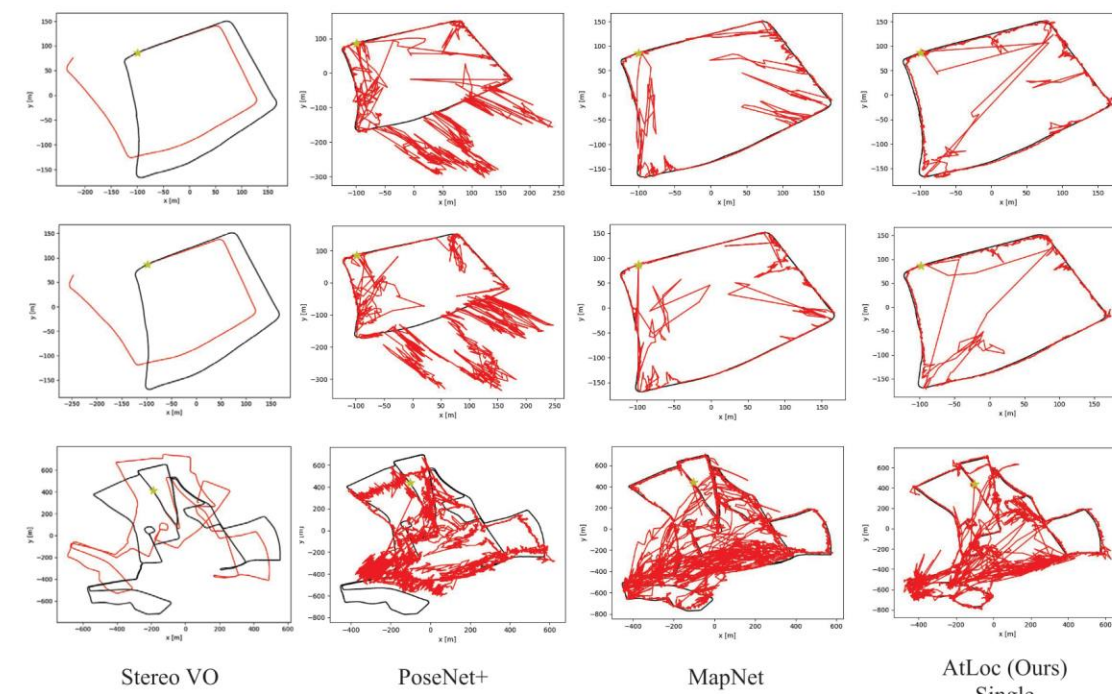
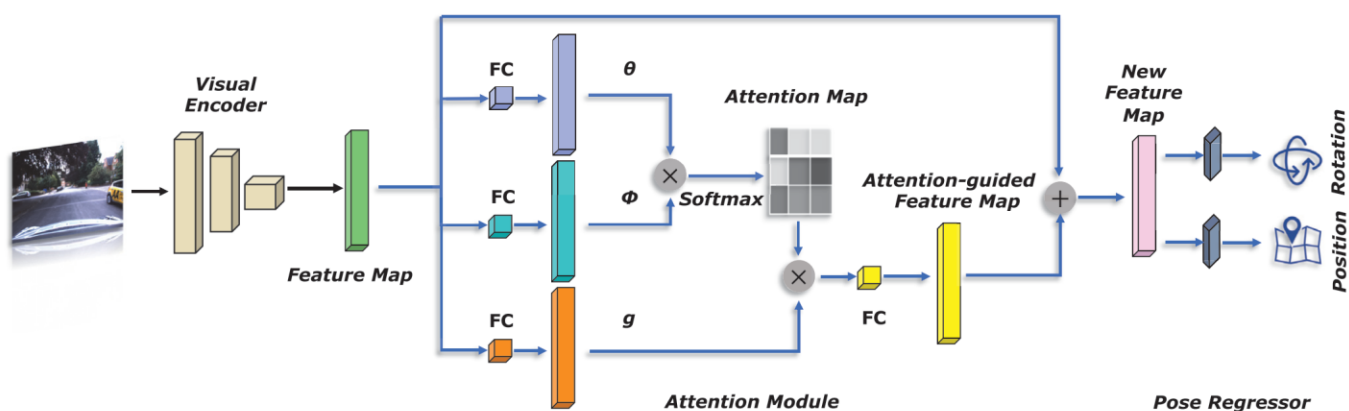


(d) Sequence length: 100.

Learning based Camera Relocalization

End-to-End Camera Relocalization

AtLoc: Attention guided camera localization (2020)



Sequence	PoseNet+		AtLoc (Ours)		MapNet		AtLoc+ (Ours)	
—	Mean	Median	Mean	Median	Mean	Median	Mean	Median
LOOP1	25.29m, 17.45°	6.88m, 2.06°	8.61m, 4.58°	5.68m, 2.23°	8.76m, 3.46°	5.79m, 1.54°	7.82m, 3.62°	4.34m, 1.92°
LOOP2	28.81m, 19.62°	5.80m, 2.05°	8.86m, 4.67°	5.05m, 2.01°	9.84m, 3.96°	4.91m, 1.67°	7.24m, 3.60°	3.78m, 2.04°
FULL1	125.6m, 27.10°	107.6m, 22.5°	29.6m, 12.4°	11.1m, 5.28°	41.4m, 12.5°	17.94m, 6.68°	21.0m, 6.15°	6.40m, 1.50°
FULL2	131.1m, 26.05°	101.8m, 20.1°	48.2m, 11.1°	12.2m, 4.63°	59.3m, 14.8°	20.04m, 6.39°	42.6m, 9.95°	7.00m, 1.48°
Average	77.70m, 22.56°	55.52m, 11.7°	23.8m, 8.19°	8.54m, 3.54°	29.8m, 8.68°	12.17m, 4.07°	19.7m, 5.83°	5.38m, 1.74°



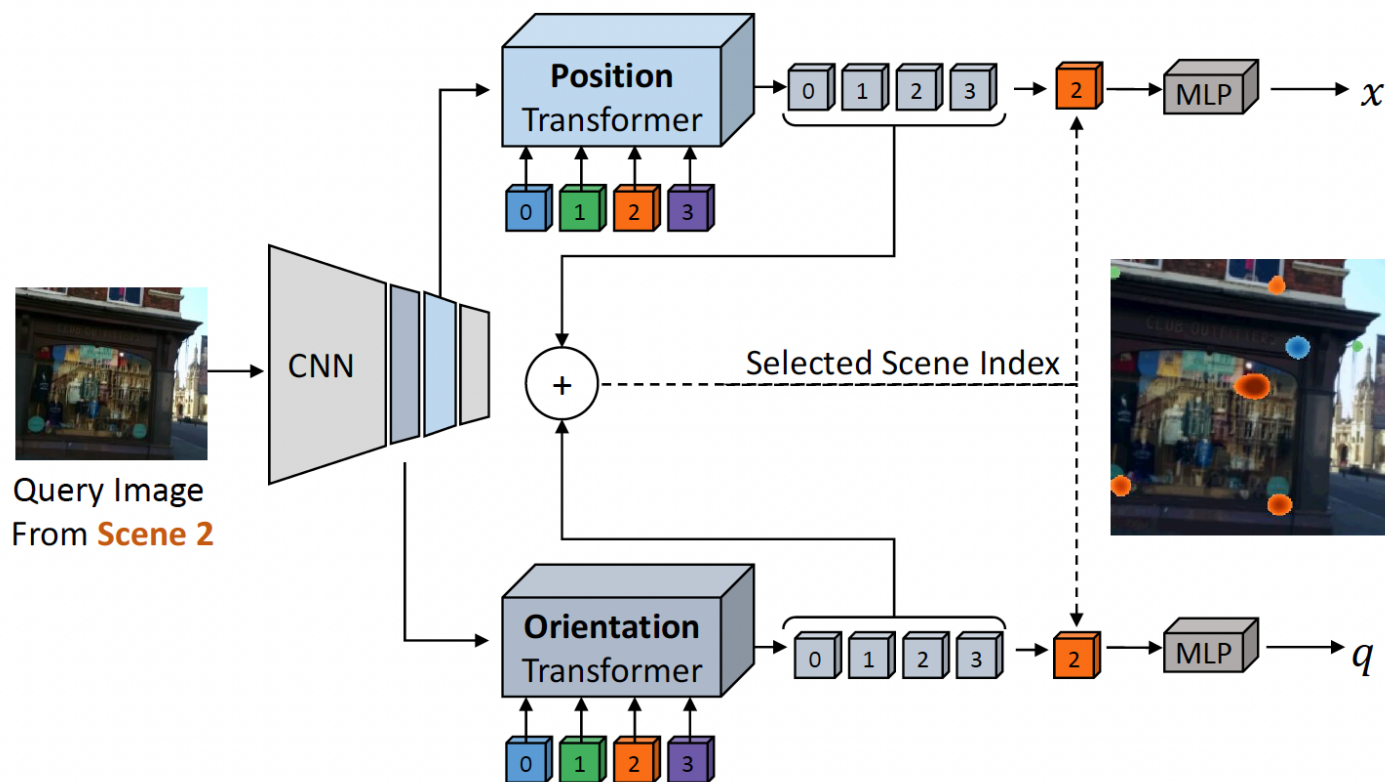
PoseNet+

AtLoc (Ours)

Learning based Camera Relocalization

End-to-End Camera Relocalization

Learning Multi-Scene Absolute Pose Regression with Transformers (2022)



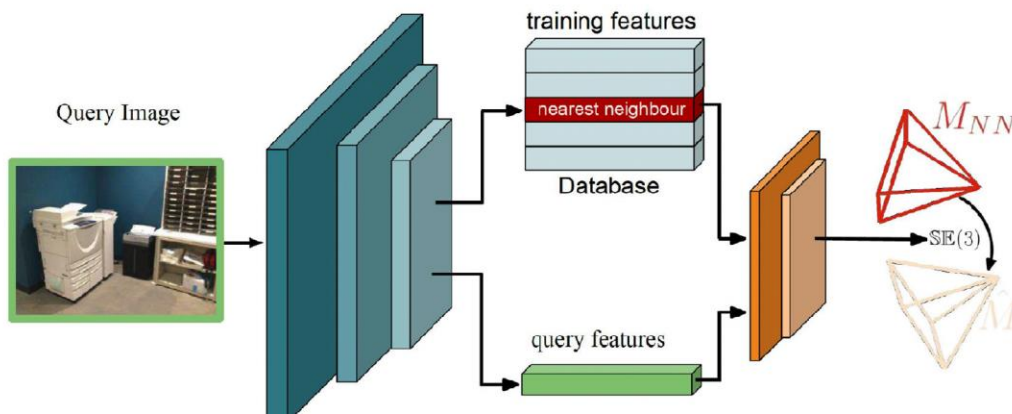
Method	Average [m/deg]	Ranks
Single-scene APRs		
PoseNet [17]	0.44/10.4	10/11
BayesianPN [15]	0.47/9.81	11/8
LSTM-PN [35]	0.31/9.86	8/9
GPoseNet [8]	0.31/9.95	8/8
PoseNet-Learnable [16]	0.24/7.87	7/4
GeoPoseNet [16]	0.23/8.12	5/5
MapNet [7]	0.21/7.78	4/3
IRPNet [29]	0.23/8.49	5/7
AttLoc [36]	0.20/7.56	2/2
Multi-scene APRs		
MSPN [3]	0.20/8.41	2/6
MS-Transformer (Ours)	0.18/ 7.28	1/1

Learning based Camera Relocalization

Map based Camera Relocalization

On the Performance of ConvNet Features for Place Recognition (2015)

- Learned visual features
- Calculate the pose from matching

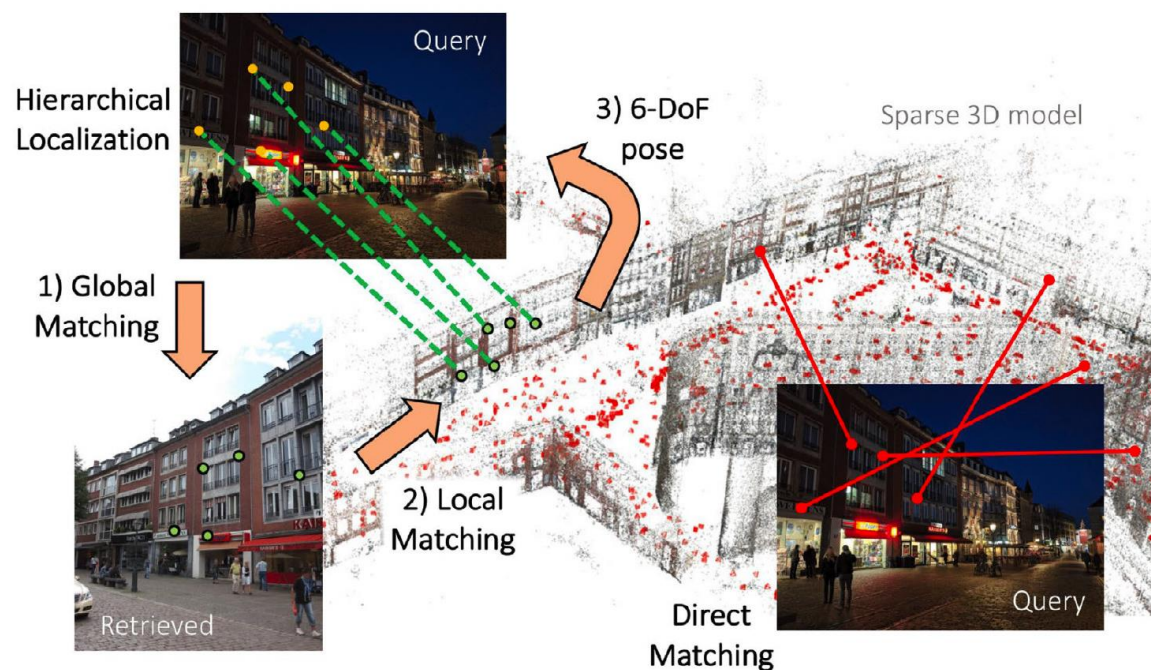


Dataset		F ₁ -Scores							SeqSLAM	FAB-MAP
		pool1	pool2	AlexNet Layers		conv5	fc6	fc7		
Nordland	Spring – Winter	0.54	0.63	0.75	0.70	0.50	0.09	0.06	0.80 *	†
	Summer – Winter	0.49	0.51	0.61	0.55	0.34	0.07	0.04	0.64 *	†
	Fall – Winter	0.52	0.60	0.66	0.60	0.38	0.07	0.05	0.63 *	†
	Spring – Summer	0.75	0.89	0.93	0.91	0.83	0.49	0.32	0.86 *	†
	Spring – Fall	0.78	0.90	0.93	0.92	0.84	0.53	0.35	0.88 *	†
Gardens Point	day left – day right	0.61	0.88	0.89	0.88	0.91	0.96	0.96	0.44	—
	day right – night right	0.66	0.89	0.94	0.93	0.92	0.82	0.73	0.44	—
	day left – night right	0.32	0.64	0.68	0.66	0.65	0.65	0.56	0.21	—
St. Lucia	average over 9 trials	—	—	0.98	0.98	0.95	0.81	0.72	—	†
	worst trial	—	—	0.96	0.96	0.87	0.51	0.38	0.59	†
Campus	human – robot	—	—	0.89	0.89	0.89	0.80	0.74	0.22	—

Learning based Camera Relocalization

Map based Camera Relocalization

From coarse to fine: Robust hierarchical localization at large scale (2019)



HFNet, a hierarchical localization approach based on a monolithic CNN that simultaneously predicts local features and global descriptors for accurate 6-DoF localization. We exploit the coarse-to-fine localization paradigm: we first perform a global retrieval to obtain location hypotheses and only later match local features within those candidate places

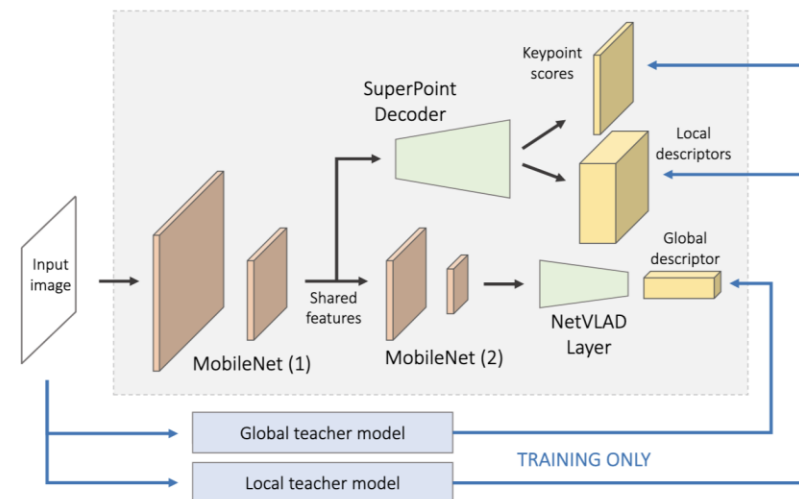
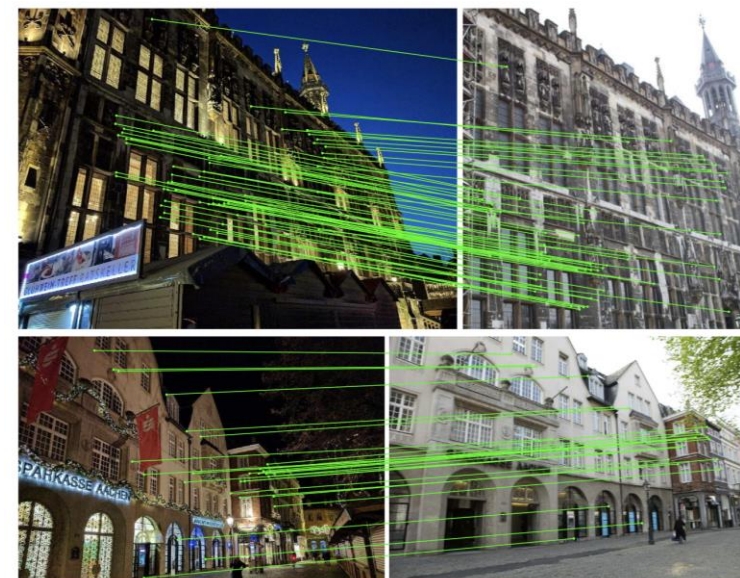


Figure 3. **HF-Net** generates three outputs from a single image: a global descriptor, a map of keypoint detection scores, and dense keypoint descriptors. All three heads are trained jointly with multi-task distillation from different teacher networks.



Learning based Camera Relocalization

Scene Coordinates Regression based Camera Relocalization

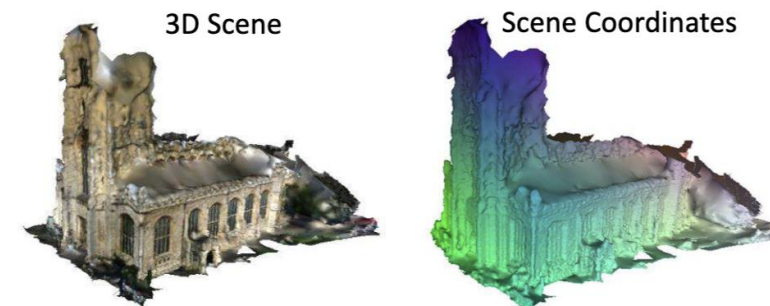
Learning less is more-6d camera localization via 3d surface regression (2018)

Table 1. **Median 6D Localization Errors.** We report results for the 7Scenes dataset [23] and the Cambridge Landmarks dataset [11]. We mark best results **bold** (if both, translational and rotational error, are lowest). Results of DSAC marked with an asterisk (*) are before end-to-end optimization which did not converge. A dash (-) indicates that a method failed completely.

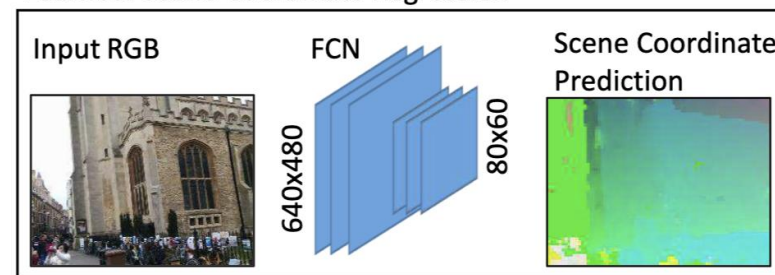
Dataset	Training w/ 3D Model				w/o 3D Model		
7Scenes	PoseNet [10] (Geom. Loss)	Active Search [20]	DSAC [2] (RGB Training)	Ours	PoseNet [10] (Pose Loss)	Spatial LSTM [29]	Ours
Chess	0.13m, 4.5°	0.04m, 2.0°	0.02m, 1.2°	0.02m, 0.5°	0.14m, 4.5°	0.24m, 5.8°	0.02m, 0.7°
Fire	0.27m, 11.3°	0.03m, 1.5°	0.04m, 1.5°	0.02m, 0.9°	0.27m, 11.8°	0.34m, 11.9°	0.03m, 1.1°
Heads	0.17m, 13.0°	0.02m, 1.5°	0.03m, 2.7°	0.01m, 0.8°	0.18m, 12.1°	0.21m, 13.7°	0.12m, 6.7°
Office	0.19m, 5.6°	0.09m, 3.6°	0.04m, 1.6°	0.03m, 0.7°	0.20m, 5.8°	0.30m, 8.1°	0.03m, 0.8°
Pumpkin	0.26m, 4.8°	0.08m, 3.1°	0.05m, 2.0°	0.04m, 1.1°	0.25m, 4.8°	0.33m, 7.0°	0.05m, 1.1°
Kitchen	0.23m, 5.4°	0.07m, 3.4°	0.05m, 2.0°	0.04m, 1.1°	0.24m, 5.5°	0.37m, 8.8°	0.05m, 1.3°
Stairs	0.35m, 12.4°	0.03m, 2.2°	1.17m, 33.1°	0.09m, 2.6°	0.37m, 10.6°	0.40m, 13.7°	0.29m, 5.1°
Cambridge							
Great Court	7.00m, 3.7°	-	*2.80m, 1.5°	0.40m, 0.2°	6.83m, 3.5°	-	0.66m, 0.4°
K. College	0.99m, 1.1°	0.42m, 0.6°	*0.30m, 0.5°	0.18m, 0.3°	0.88m, 1.0°	0.99m, 1.0°	0.23m, 0.4°
Old Hospital	2.17m, 2.9°	0.44m, 1.0°	0.33m, 0.6°	0.20m, 0.3°	3.20m, 3.3°	1.51m, 4.3°	0.24m, 0.5°
Shop Facade	1.05m, 4.0°	0.12m, 0.4°	0.09m, 0.4°	0.06m, 0.3°	0.88m, 3.8°	1.18m, 7.4°	0.09m, 0.4°
St M. Church	1.49m, 3.4°	0.19m, 0.5°	*0.55m, 1.6°	0.13m, 0.4°	1.57m, 3.2°	1.52m, 6.7°	0.20m, 0.7°
Street	20.7m, 25.7°	0.85m, 0.8°	-	-	20.3m, 25.5°	-	-

Predict, for every image pixel, its corresponding 3D scene coordinate in a global reference frame.

Once pixel → 3D correspondences are known, the camera pose is recovered via PnP + RANSAC.



Learned: Scene Coordinate Regression



Fixed but Differentiable: Pose Optimization



Figure 2. **System Overview.** Given an RGB image, we estimate the 6D camera pose in two stages. Firstly, a fully convolutional network (FCN) regresses 3D scene coordinates (XYZ mapped to RGB for visualization). This CNN is the only learnable component of our system. Secondly, the system optimizes the pose by sampling a pool of hypotheses, scoring them using a soft inlier count, selecting one according to the scores, and refining it as the final estimate. The second stage contains no learnable parameters but is fully differentiable.

Summary

Advantages

- Robust to Appearance Changes**

Learns invariances to illumination, weather, season, and viewpoint changes

- No Need for Hand-Crafted Features**

Deep features outperform SIFT/ORB under challenging conditions.

- Scalable Retrieval-Based Localization**

Efficient global place recognition in large-scale environments.

Key Challenges

- Generalization Across Environments**

Models trained on one city/season/domain often fail in new lighting, weather, or structural conditions.

- Scene-Specific Training Requirements**

Scene coordinate methods need a pre-built 3D map; adaptation to new scenes is costly.

- Dependence on Large-Scale Data**

High-quality ground-truth poses and dense mapping data are expensive to obtain.

- Domain Shift & Long-Term Robustness**

Day–night, seasonal changes, occlusions, and dynamic objects degrade localization reliability.



Thanks for your attention!

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